

DSC2F: An Multimodal AI-Generated Image Detection System Using Dual-Stream Convolutional Neural Networks With Fast Fourier Transform

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Abstract. The rapid advancement of generative models, including GANs and diffusion-based architectures, has made AI-generated images increasingly indistinguishable from real photographs, posing critical challenges for digital forensics and media verification. This paper proposes a Dual-Stream CNN detection system, called DualStreamCNNEnhanced or DSCNNE, that processes images through two parallel branches: a Spatial Stream analyzing RGB features and a Frequency Stream analyzing the FFT amplitude spectrum. A Cross-Modal Attention mechanism fuses the two 512-dimensional feature vectors before a Binary Cross-Entropy classifier produces the final prediction. The enhanced model integrates 30 SRM high-pass filters, CBAM attention modules, SE Blocks, and a ResNet-style backbone, totaling approximately 7.8M parameters.

Both the baseline (DualStreamCNN, 3.7M parameters) and the proposed model are trained on CIFAKE (120,000 images) and evaluated zero-shot on the ArtiFact dataset (20,000 images, 13+ generators). Zero-shot evaluation reveals near-chance performance across all models (accuracy 49–51%), exposing a critical generalization gap from single-generator training to multi-generator evaluation. Our Algorithms achieves the most balanced predictions with AccF = 53.75% and AUC = 51.01%, and reduces false alarms on real images compared to the baseline. These results highlight the

critical importance of multi-generator training for robust real-world deployment, and motivate future work on training with diverse datasets such as ArtiFact and GenImage.

Keywords: AI Image Detection, Dual-Stream CNN, Frequency Analysis, Attention Mechanism, Digital Forensics.

INTRODUCTION

The rapid advancement of image synthesis models based on Generative Adversarial Networks (GANs) and Diffusion Models, including ProGAN, StyleGAN, Stable Diffusion, DALL-E 3, and Midjourney, has created serious challenges in verifying the authenticity of digital images. The quality of AI-synthesized imagery now exceeds the threshold of human visual discrimination, creating an urgent need for reliable detection tools in media, legal, and national security contexts.

The passive detection approach, which analyzes pixel content directly without relying on image metadata, offers significant advantages in independence and offline deployability. The proposed Our Algorithms selectively inherits from four research directions (spatial domain, frequency domain, attention mechanisms, backbone architecture) and adds a zero-shot evaluation strategy on the ArtiFact dataset, representing the most challenging and realistic scenario for detecting AI content across diverse generators.

Dual-Stream Architecture with Cross-Modal Attention: We propose DSCNNE, an enhanced detection framework that effectively integrates spatial pixel-level features with frequency-domain artifacts (FFT amplitude spectrum). A bidirectional Cross-Modal Attention mechanism is introduced to dynamically fuse features from both streams, allowing the model to focus on the most discriminative signals for AI detection.

Integration of SRM and Attention Modules: We enhance the standard CNN backbone by incorporating 30 Steganalysis Rich Model (SRM) high-pass filters to amplify micro-noise patterns and Convolutional Block Attention Modules (CBAM) to refine spatial and channel-wise feature representation. This combination significantly improves the

detection of subtle GAN and diffusion-based artifacts.

Robust Empirical Evaluation and Generalization Analysis: We provide a comprehensive zero-shot evaluation on the ArtiFact benchmark (Awsaf et al., 2023), a large-scale dataset covering 13 or more generators across five semantic categories. All models achieve near-chance performance (accuracy 49–51%), exposing a critical generalization gap from single-generator CIFAKE training to diverse

real-world generators. This negative result is itself a meaningful contribution, highlighting the limitations of current detection approaches. **Real-world Generalization Gap Analysis:** Unlike many existing studies that evaluate on the same distribution as training, we test zero-shot on ArtiFact, which contains images from generators not seen during training. The near-chance AUC across all models (49–51%) confirms that retraining on multi-generator datasets is essential for practical deployment.

RELATED WORK

The field of AI-generated image detection has evolved in parallel with the pace of image synthesis model development. Two fundamental approaches exist: passive detection and active provenance verification. On the passive detection side, commercial SaaS tools such as Hype Lense, AI or Not, Illuminarty, and Hive Moderation analyze image pixel content directly. On the active side, Adobe implements the Content Authenticity Initiative (CAI) following the C2PA standard to embed cryptographically signed metadata into image files, while Google SynthID invisibly embeds a digital watermark into pixels of images generated by Imagen.

Wang et al. (2020) established the most important foundation for GAN image detection: a simple CNN binary classifier, trained exclusively on ProGAN images, generalizes remarkably to dozens of unseen GAN architectures (StyleGAN, BigGAN, CycleGAN, StarGAN) with average AUC exceeding 99%, owing to a data augmentation strategy combining Gaussian blur and random JPEG compression. Zhang et al. (2019) proposed AutoGAN to simulate and detect artifacts characteristic of the standard GAN

pipeline, demonstrating that a general-purpose detector can be built by exploiting common features of the AI synthesis process rather than building generator-specific detectors.

Frank et al. (2020) directly influenced the design of the Frequency Stream. The authors discovered that GAN images contain clear artifacts in the frequency domain that are invisible to the human eye: the transposed convolution operation, widely used in GAN decoders for upsampling, produces a checkerboard pattern in the frequency spectrum as periodically spaced energy peaks, while real

images exhibit a monotonically decreasing spectrum from center to periphery. Frank et al. (2020) built a classifier based solely on the FFT spectrum and achieved high AUC across multiple GAN architectures, proving that the frequency domain provides discriminative signals that are supplementary to and independent of the spatial domain, forming the direct theoretical basis for the FFT transformation formulas in the proposed system.

Qian et al. (2020) extended frequency-domain analysis to face deepfake detection in Thinking in Frequency. The authors proposed the F3Net module, using the Discrete Cosine Transform (DCT) instead of FFT, exploiting two types of features: frequency-aware decomposed image components and local frequency statistics, combined via a cross-attention mechanism to detect manipulated facial regions. Nataraj et al. (2019) proposed a distinct approach: using co-occurrence matrices directly on RGB pixel channels to capture statistical correlations between neighboring pixels, exploiting differences in pixel correlation structure between real and GAN images since the GAN synthesis process tends to produce unnatural pixel correlation patterns compared to real photographs.

The emergence of Denoising Diffusion Probabilistic Models or DDPM (Ho et al., 2020) and their improved variants (Dhariwal & Nichol, 2021; Rombach et al., 2022; Song et al., 2021) has disrupted the entire AI image detection field. Unlike GANs, which leave clear checkerboard patterns in the frequency domain, diffusion models synthesize images through an iterative denoising process, producing frequency spectra that are more natural and significantly harder to distinguish. Ho et al. (2020) introduced DDPM with forward and reverse diffusion mechanisms; Song et al. (2021) proposed DDIM enabling faster synthesis without quality loss; and Rombach et al. (2022) advanced diffusion models further with Latent Diffusion Models, the technical basis of Stable Diffusion, enabling high-resolution image synthesis at reasonable

computational cost.

He et al. (2016) resolved the accuracy degradation problem with increasing network depth through skip connections in ResNet, enabling training of networks hundreds of layers deep. Woo et al. (2018) proposed CBAM, a lightweight attention module insertable

into any CNN architecture, operates in two sequential stages: channel attention identifies the most important feature channels by combining Global Average Pooling and Global Max Pooling through an MLP; followed by spatial attention that identifies the spatial regions requiring focus by convolving over the combination of two channel-wise pooling operations.

The CNN Detection system selectively inherits from all four research directions: dual-stream architecture combining spatial analysis (Wang et al., 2020) with frequency analysis (Frank et al., 2020); SRM filters for pixel-level artifact detection; CBAM (Woo

et al., 2018) for channel and spatial focusing; and Cross-Modal Attention to fuse cross-modal information. The key differentiator is the supplementary evaluation strategy on purely real CIFAR-10 images to measure FPR in the most realistic deployment scenario.

PROPOSED ALGORITHMS

Our algorithms, called DualStreamCNNEnhanced or DSCNNE, is built on the Dual-Stream CNN architecture, processing images in parallel through two independent branches: the Spatial Stream analyzes RGB spatial features, and the Frequency Stream analyzes the FFT amplitude spectrum. Two 512-dimensional feature vectors are fused via Cross-Modal Attention before the MLP classifier. our algorithm adds over the baseline: 30 SRM high-pass filters, CBAM modules, SE Blocks inside EnhancedResidualBlocks, and a ResNet-enhanced backbone, totaling $\sim 7.8\text{M}$ parameters.

Fig. 1. Overview architecture diagram of our algorithm.

Figure 1 illustrates the overall processing pipeline of our algorithm. An input image is first converted to grayscale and processed by 30 SRM (Steganalysis Rich Model) high-pass filters to extract noise residuals.

Simultaneously, a 2D Fast Fourier Transform (FFT) produces the frequency-domain amplitude spectrum. The Spatial Stream (left branch) feeds RGB and SRM features through an EnhancedResidualBlock backbone with CBAM attention, producing

a 512-dimensional spatial feature vector. The

Frequency Stream (right branch) processes the log-scale FFT amplitude through a parallel CNN backbone, also yielding a 512-dimensional frequency feature vector. A bidirectional Cross-Modal Attention module then fuses both streams, allowing each branch to selectively weight the most informative signals from the other. The two 512-dimensional attended vectors are concatenated into a 1024-dimensional joint representation, which is fed to an MLP classifier that outputs a binary prediction: real (0) or AI-generated fake (1).

Grayscale conversion.

$$G = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B \quad (1)$$

where G is the grayscale output; R, G, B are the red, green, and blue channel pixel values; the coefficients 0.299, 0.587, and 0.114 are the standard luminance weights (ITU-R BT.601).

2D discrete Fourier transform.

$$F_{u,v} = x(y) \cdot \exp\left\{j2\pi \left[\frac{u}{W} \left(x - \frac{W}{2} \right) + \frac{v}{H} \left(y - \frac{H}{2} \right) \right] \right\} \quad (2)$$

Log-scale amplitude spectrum.

$$M = \log(1 + |F|) \quad (3)$$

where $A(u,v)$ is the log-scale amplitude at frequency (u,v) ; $F(u,v)$ is the complex DFT coefficient; the $\log(1+\cdot)$ operation compresses the dynamic range, making subtle high-frequency artifacts more visible for the subsequent CNN feature extractor.

The Spatial Stream (EnhancedSpatialStream) processes RGB input through a Stem Block (Conv 7×7 /stride=2, BN, ReLU, MaxPool) then through three EnhancedResidualBlocks with increasing channel counts $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$, each followed by CBAM. Each EnhancedResidualBlock integrates an SE Block (reduction=16) after

the second convolution, with a 1×1 Conv skip connection when dimensions change. Output: $F_{\text{spatial}} \in \mathbb{R}^{(B \times 512)}$ via Global Average Pooling and Flatten.

The Frequency Stream (EnhancedFrequencyStream) extracts multi-scale features from the single-channel FFT spectrum through three parallel branches: Conv 7×7 (low-frequency, global artifacts), Conv 5×5 (mid-frequency), and Conv 3×3 (high-frequency, local detail), each producing 32 channels. The three outputs are concatenated to 96 channels, then processed through three Conv $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ CBAM blocks ($96 \rightarrow 128 \rightarrow 256 \rightarrow 512$). This multi-scale design simultaneously captures global and local frequency structure. Output: $F_{freq} \in \mathbb{R}(B \times 512)$.

Channel Attention recalibrates X along the channel dimension:

$$\alpha = \text{Sigmoid}(\text{FC}_2 \text{ReLU}(\text{FC}_1 \text{AvgPool} X + \text{FC}_2 \text{ReLU}(\text{FC}_1 \text{MaxPool} X))) \quad (4)$$

where $\alpha \in \mathbb{R}^C$ is the channel attention vector; z_{avg} and z_{max} are global average- and max-pooled feature vectors over the spatial dimensions; FC_1 and FC_2 are fully-connected layers with a bottleneck reduction ratio $r=16$; ReLU is the activation; and σ is the sigmoid function.

Spatial Attention recalibrates X along spatial positions:

$$\beta = \text{Sigmoid}(\text{Conv}_{7 \times 7}(\text{MeanChannel} X; \text{MaxChannel} X)) \quad (5)$$

where $\beta \in \mathbb{R}^{(H \times W)}$ is the spatial attention map; MeanChannel and MaxChannel apply channel-wise average and max pooling to produce two spatial maps; $\text{Conv}_{7 \times 7}$ is a 7×7 convolution; and σ is the sigmoid function. The output $X_{out} = \beta \otimes X$ recalibrates each spatial location.

Spatial attends to Frequency:

$$Q_s = \text{Linear} F_{spatial}, K_f = \text{Linear} F_{freq}, V_f = \text{Linear} F_{freq}$$

$$\text{attn} = \text{Sigmoid}(Q_s \times K_f^T) F_{spatial_enh} = F_{spatial} + \text{attn} \times V_f \quad (6)$$

where Q_s , K_f , V_f are query, key, and value projections of the spatial and frequency features via learned linear layers; d_k is the key dimension used for scaling; $\text{softmax}(\cdot)$ produces attention weights over the frequency feature sequence; and $F_{\text{spatial_enh}}$ is the enhanced spatial representation after cross-modal attention.

Frequency attends to Spatial symmetrically. Residual connections ensure training stability. The two enhanced vectors are concatenated.

$$F_{\text{combined}} = [F_{\text{spatial_enh}}; F_{\text{freq_enh}}] \in \mathbb{R}^{B \times 1024} \quad (7)$$

where $F_{\text{spatial_enh}}$ and $F_{\text{freq_enh}}$ are the cross-attention-enhanced spatial and frequency representations (each $\in \mathbb{R}^{B \times 512}$); $[\cdot; \cdot]$ denotes channel-wise concatenation; and $F_{\text{combined}} \in \mathbb{R}^{B \times 1024}$ is the joint multimodal representation fed to the MLP classifier.

The MLP Classifier processes F_{combined} through the following layers.

$$\text{Linear}(1024 \rightarrow 1024) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.5) \quad (8)$$

First Compression Layer.

$$\text{Linear}(1024 \rightarrow 512) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.5) \quad (9)$$

Second Compression Layer.

$$\text{Linear}(512 \rightarrow 256) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.25) \quad (10)$$

Output Layer.

$$\text{Linear}(256 \rightarrow 1) \rightarrow \text{logit } Y \quad (11)$$

At inference.

$$\text{Sigmoid}(Y) \rightarrow P(\text{FAKE}) \in [0, 1] \quad (12)$$

where $\hat{y} = \sigma(Y)$ is the predicted probability; Y is the raw logit output of the classifier; σ is the sigmoid function; and the threshold 0.5 maps probabilities to binary labels (1 = AI-generated fake, 0 = real).

The loss function for optimization is Binary Cross-Entropy.

$$LBCE = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(y_i) + (1 - y_i) \cdot \log(1 - y_i)] \quad (13)$$

where L_{BCE} is the Binary Cross-Entropy loss; N is the number of training samples; $y_i \in \{0,1\}$ is the ground-truth label (0 = real, 1 = fake); $y_i = \sigma(y_i) \in (0,1)$ is the predicted probability after sigmoid activation; and $\log(\cdot)$ is the natural logarithm.

The pseudocode of our algorithm, called our algorithm (DSCNNE), is described below:

ALGORITHM our algorithm(X_{rgb} , X_{fft})

Input : X_{rgb} in $R^{(B \times 3 \times 224 \times 224)}$ -- RGB image

X_{fft} in $R^{(B \times 1 \times 224 \times 224)}$ -- FFT spectrum (from

ComputeFFTSpectrum)

Output: Y in $R^{(B \times 1)}$ -- Logit (Real=1 / Fake=0)

BEGIN

// Step 1: SRM filter bank (30 high-pass filters)

$X_{srm} \leftarrow \text{ApplySRMFilters}(X_{rgb}, k=30)$ // $R^{(B \times 30 \times H \times W)}$ $X_{in} \leftarrow$

$\text{Concat}([X_{rgb}, X_{srm}], \text{dim}=\text{channel})$ // $R^{(B \times 33 \times H \times W)}$

// Step 2: Dual-stream feature extraction

F_spatial <- EnhancedSpatialStream(X_in) // $R^{(B \times 512)}$

F_freq <- EnhancedFrequencyStream(X_fft) // $R^{(B \times 512)}$

// Step 3: Cross-Modal Attention (bidirectional)

Q_s <- Linear_Qs(F_spatial); K_f <- Linear_Kf(F_freq); V_f <-
Linear_Vf(F_freq)

attn_s <- Sigmoid(Sum(Q_s * K_f, dim=-1) / sqrt(512))

F_spatial_e <- F_spatial + attn_s * V_f // residual

```

Q_f <- Linear_Qf(F_freq); K_s <- Linear_Ks(F_spatial); V_s <-
Linear_Vs(F_spatial)

attn_f <- Sigmoid(Sum(Q_f * K_s, dim=-1) / sqrt(512))

F_freq_e <- F_freq + attn_f * V_s // residual

// Step 4: Feature fusion

F_combined <- Concat([F_spatial_e, F_freq_e], dim=1) //  $R^{(B \times 1024)}$ 

// Step 5: MLP Classifier

Z <- Dropout(ReLU(BN(Linear(1024->1024)(F_combined)))),
p=0.50)

Z <- Dropout(ReLU(BN(Linear(1024->512 )(Z))), p=0.50)

Z <- Dropout(ReLU(BN(Linear(512 ->256 )(Z))), p=0.25)

Y <- Linear(256->1)(Z)

RETURN Y // Sigmoid(Y) at inference -> P(FAKE) in [0,1]

END

```

EXPERIMENTS

Benchmark. Following training, a zero-shot generalization evaluation was performed on the ArtiFact dataset (Awsaf et al., 2023), a large-scale benchmark containing images generated by 13 or more generators across five semantic categories (Human/Faces, Animal, Places, Vehicles, and Art), originally used in the IEEE VIP Cup at ICIP 2022. A balanced subset of 20,000 images (10,000 real, 10,000 AI-

generated) was used.

Experimental Setup. To evaluate the proposed architecture, experiments were conducted on a Windows 11 workstation

equipped with an Intel Core i7-11800H CPU, 16 GB DDR4 RAM, and an NVIDIA RTX 3050Ti GPU (4 GB VRAM), utilizing Python 3.10, PyTorch 2.0.1, and CUDA 11.7. Both the proposed and baseline models were trained from scratch on the CIFAKE dataset (Bird & Lotfi, 2024), which comprises 120,000 real CIFAR-10 photographs and AI-generated images from Stable Diffusion v1.4. Prior to training, input images were preprocessed via a 224×224 center-crop and standard ImageNet normalization ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). The optimization process employed the Adam optimizer ($\text{lr} = 0.0001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) with a batch size of 16. To ensure optimal convergence and prevent overfitting, we incorporated a ReduceLROnPlateau scheduler (factor = 0.1, patience = 5 epochs) alongside an Early Stopping mechanism (patience = 10 epochs). All evaluations were executed on the CPU to ensure a fair and hardware-agnostic speed comparison.

The primary evaluation metric is Average Precision (AP), which is threshold-independent and robust to class imbalance. Models were also assessed on overall accuracy (Acc), accuracy on real (AccR) and fake (AccF) images, F1 score on fake images, and total evaluation time. Metric definitions: AP (Average Precision) measures the area under the precision-recall curve, summarizing detection quality independently of any fixed threshold; Acc (Overall Accuracy) is the fraction of all predictions that are correct; AccR (Accuracy on Real) is the fraction of real images correctly classified as real (true-negative rate); AccF (Accuracy on Fake) is the fraction of fake images correctly classified as fake (true-positive rate); F1 is the harmonic mean of precision and recall on the fake class.

Results and Discussion. Table 1 presents the zero-shot evaluation results of all three models on the 20,000-image ArtiFact dataset (10,000 real, 10,000 AI-generated from 13+ generators). All three models perform near chance level, with accuracy ranging from 48.74% to 50.91%, indicating poor generalization to this highly diverse out-of-distribution benchmark.

Table 1. Comparison of all models on ArtiFact dataset (20,000 images, zero-shot).

Metric

Wang et al. (2020)

(Baseline)

Our Algorithms

AP (Average Precision)

50.32%

51.01%

Acc (Overall Accuracy)

48.74%

50.91%

AccR (Accuracy on Real)

18.61%

48.08%

AccF (Accuracy on Fake)

78.88%

53.75%

AUC (ROC-AUC)

49.13%

51.01%

F1 (Fake Class)

60.61%

52.27%

Evaluation Time (20,000 images)

49.1s

52.4s

Parameter Count

3.7M

7.8M

Metric definitions AP (Average Precision): area under the precision-recall curve, threshold-independent; Acc (Overall Accuracy): fraction of all predictions that are correct; AccR (Accuracy on Real): fraction of real images correctly classified as real (true-negative rate); AccF (Accuracy on Fake): fraction of fake images correctly classified as fake (true-positive rate); AUC (ROC-AUC): area under the receiver operating characteristic curve; F1 (Fake Class): harmonic mean of precision and recall for the fake class; Evaluation Time: total CPU inference time on 20,000 images; Parameter Count: total trainable model parameters.

On the ArtiFact benchmark, all three models perform near random chance. Our Algorithms achieves the highest accuracy (50.91%) and AUC (0.51), slightly outperforming DualStreamCNN (48.74%) and FFTOnlyCNNEnhanced (49.60%), but the differences are marginal.

This near-chance performance across all models indicates that features learned from CIFAKE (Stable Diffusion v1.4 images at 32×32

resolution) do not generalize to the diverse, multi-generator, high-resolution content of ArtiFact. The result highlights a critical limitation of training on a single-generator dataset for real-world deployment.

Notably, the three models show different classification biases on

ArtiFact. DualStreamCNN exhibits a strong FAKE bias: AccR = 18.61% versus AccF = 78.88%, suggesting it classifies most images as fake regardless of content. Our Algorithms shows a more balanced split (AccR = 48.08%, AccF = 53.75%), indicating its CBAM and cross-modal attention provide some generalization benefit. Table 2 compares the three models on this AccR vs. AccF trade-off.

Table 2. Trade-off between AccR (real images) and AccF (fake images) on ArtiFact (zero-shot).

Model

AccR (Real)

AccF (Fake)

Overall Acc

AP

Wang et al. (2020)

18.61%

78.88%

48.74%

50.32%

Our Algorithms

48.08%

53.75%

50.91%

51.01%

our algorithm pushes the FAKE bias further: AccR increases from 18.61% to 48.08%, while AccF decreases from 78.88% to 53.75%. This shows that the attention mechanisms in Our Algorithms reduce the extreme FAKE bias of the baseline, producing a more calibrated detector — though both models still operate near random chance overall, reflecting the generalization gap from single-generator CIFAKE training to multi-generator ArtiFact evaluation.

Table 3 provides a full summary of all three model variants evaluated on the ArtiFact benchmark, including evaluation time. The FFTOnlyCNNEnhanced model, which uses only the frequency stream, takes the least time (41.5s) but shows no accuracy advantage over the dual-stream models, confirming that frequency features alone are insufficient for generalizing across the diverse generators and image categories in ArtiFact.

Table 3. Full comparison of all three model variants on ArtiFact (20,000 images, zero-shot).

Model

Acc (%)

AUC

AP (%)

F1 (%)

Prec (%)

Rec (%)

Time (s)

Wang et al. (2020)

48.74

49.13

50.32

60.61

49.22

78.88

49.1

Our algorithms (Single-modal FFT)

49.60

50.25

50.66

57.72

49.71

68.79

41.5

Our Algorithms (Multi-modal FFT& CNN)

50.91

51.01

51.01

52.27

50.87

53.75

52.4

Table 4 provides the confusion matrix breakdown for all three models on the ArtiFact dataset, showing the number of True Negatives (TN, real images correctly classified), False Positives (FP, real images wrongly flagged as fake), False Negatives (FN, fake images missed), and True Positives (TP, fake images correctly detected). DualStreamCNN has the highest TP count but at the cost of many FP errors on real images, while Our Algorithms achieves the most balanced prediction across both classes.

Table 4. Confusion matrix breakdown on ArtiFact (10,000 real + 10,000 fake images).

Model
TN (Real Correct)
FP (Real Wrong)
FN (Fake Missed)
TP (Fake Correct)
DualStreamCNN
1,861
8,139
2,112
7,888
Our Algorithms

4,808

5,192

4,625

5,375

FFTOnlyCNNEnhanced

3,042

6,958

3,121

6,879

As shown in the confusion matrix, our algorithms classifies 4,808 real images correctly (TN) versus only 1,861 for DualStreamCNN, a gain of 2,947 additional correctly classified real images, at the cost of a more balanced but lower TP count (5,375 vs. 7,888). This trade-off reflects the effect of CBAM and Cross-Modal Attention in reducing the strong FAKE bias present in the baseline. While none of the models generalize well to ArtiFact’s diverse generators in terms of overall accuracy, Our Algorithms produces the most balanced predictions, making it better suited for scenarios where false alarms on real images are costly..

The generalization gap observed on ArtiFact reveals an important limitation of single-generator training. SRM filters amplify pixel-level micro-noise and GAN checkerboard artifacts (Frank et al., 2020), while CBAM (Woo et al., 2018) directs the model’s attention to artifact-rich regions. These components are effective for detecting Stable

Diffusion v1.4 artifacts (the CIFAKE training distribution), but ArtiFact images from 13 or more generators introduce diverse artifact signatures that the models have not encountered during training. The near-chance AUC (49–51%) across all models confirms this is a distribution shift problem rather than a model capacity problem, motivating future retraining on multi-generator datasets such as

GenImage or ArtiFact itself., making our algorithm both more accurate and better suited for production deployment.

CONCLUSION

This paper has presented Our Algorithms (DSCNNE), a Dual-Stream CNN architecture that combines spatial analysis (RGB and SRM) with frequency analysis (FFT), incorporating a Cross-Modal Attention mechanism and a Binary Cross-Entropy classifier. Both models were trained on CIFAKE and evaluated zero-shot on the ArtiFact benchmark (20,000 images, 13+ generators). All models perform near chance level on ArtiFact (accuracy 49–51%), revealing a significant generalization gap when moving from single-generator to multi-generator evaluation. Our Algorithms achieves the most balanced predictions ($\text{AccF} = 53.75\%$, $\text{AUC} = 51.01\%$) with fewer false alarms on real images than the baseline. These results indicate the need for multi-generator training to achieve reliable real-world deployment.

Three improvement directions are proposed for the next phase. The first is retraining on multi-generator datasets such as ArtiFact or GenImage (Zhu et al., 2023), which cover diverse generators and image categories, directly addressing the generalization gap identified in this evaluation. The second is integrating DIRE (Wang et al., 2023) as a third stream in the architecture, exploiting diffusion model reconstruction error to supply detection signals not currently captured by the RGB and FFT streams. The third is applying few-shot or continual learning techniques to enable the model to adapt incrementally as new AI image generators emerge, without requiring full dataset recollection., combined with a continual learning mechanism to enable the model to automatically update as new AI image generators continue to emerge.

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